

Report on Sentiment Analysis Using the Stanford IMDB Dataset

1. Title

Sentiment Analysis of Movie Reviews Using Deep Learning Models

2. Introduction

The goal of this project was to classify movie reviews from the Stanford IMDB Large Movie Review Dataset into positive or negative sentiment. The dataset contains 50,000 labeled movie reviews split evenly between training and testing sets. This makes it a strong benchmark for binary sentiment analysis using deep learning.

3. Objective

The main objective was to build and compare multiple neural network models for sentiment classification. The project focused on text preprocessing, tokenization, sequence padding, model training, and evaluation using common classification metrics.

4. Dataset Description

The Stanford IMDB dataset is organized into `train/pos`, `train/neg`, `test/pos`, and `test/neg` folders after extraction. Each review is stored as a text file, and the labels are binary: positive for favorable reviews and negative for unfavorable reviews.

5. Methodology

The project followed a structured pipeline. First, the raw review text was loaded from the extracted dataset folders. Then the text was cleaned by lowercasing and removing unnecessary symbols. After that, a tokenizer was used to convert words into integer sequences, and all sequences were padded to a fixed length so they could be passed into neural networks.

Three deep learning architectures were used:

- 1D CNN.
- BiLSTM.
- GRU.

These models were trained on the training split and validated on a separate validation set. Early stopping and model checkpointing were used to retain the best-performing model during training.

6. Results

The models were evaluated using accuracy, precision, recall, and F1-score. A confusion matrix was also generated to analyze correct and incorrect predictions. Model comparison showed which architecture handled the sentiment classification task best on the test set.

7. Discussion

The deep learning models were able to learn useful patterns from the review text and distinguish positive from negative sentiment. In general, recurrent models such as BiLSTM and GRU are better suited for sequence understanding, while CNNs often train faster and can still perform strongly on text classification tasks. The final model choice depended on the balance between performance and training efficiency.

8. Conclusion

This project demonstrated that the Stanford IMDB dataset can be used effectively for sentiment analysis with deep learning. By combining proper preprocessing, tokenization, padding, and model comparison, the project achieved a complete text classification workflow. The work also showed that careful evaluation is necessary to select the best model rather than relying only on training accuracy.

9. Future Work

Possible improvements include using pretrained embeddings such as GloVe, adding an attention mechanism, tuning hyperparameters further, and deploying the model in a small web application for live movie review prediction.